

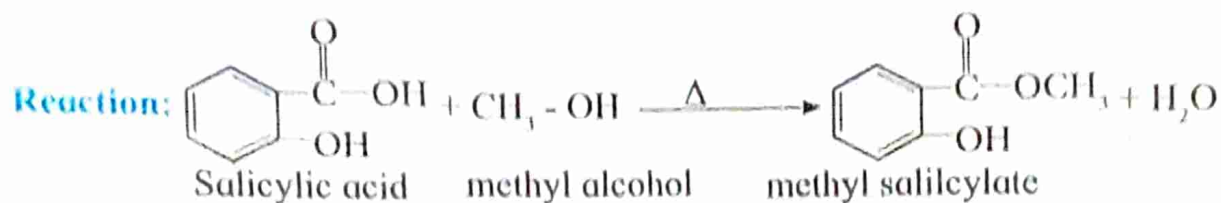
Activity No. 3

Date: / /

Aim: To prepare methyl Salicylate (oil of winter green) qualitatively.

Theory:

Carboxylic acids are converted to ester by the Fischer esterification, an acid catalysed reaction with an alcohol.



Procedure:

1. Take a small amount of salicylic acid in a hard glass test tube.
2. Add 2-3 mL of methyl alcohol.
3. Add 1 mL concentrated H_2SO_4 .
4. Heat the test tube for 10 to 15 minutes at low flame.
5. Observe the colour and take odour of the formed compound.

Observation and Result :

1. Colour of the compound = Colourless
2. Odour of the compound = Sweet, fruity test.

Remark and sign of teacher:

MCQ

Select the correct answer for the following multiple choice type questions.

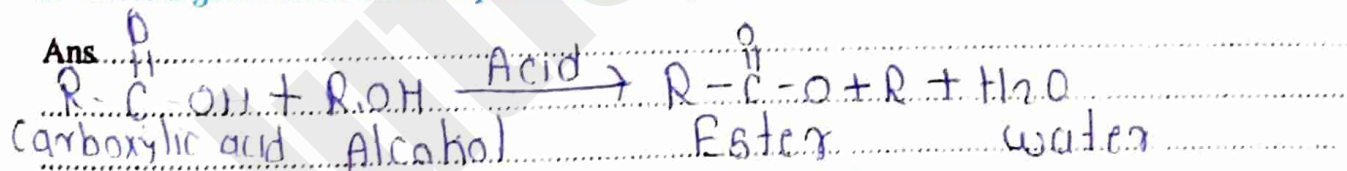
- The molecular formula of methyl salicylate is.....
 a. $C_8H_7O_3$ ☒ b. $C_8H_8O_3$ c. $C_7H_7O_3$ d. $C_8H_{10}O_3$
- The hybridisation of carbon atoms of methyl salicylate is.....
 a. $SP-SP^2$ b. SP^2-SP ☒ c. SP^3-SP^2 d. SP^3-SP
- Another name of methyl salicylate is.....
 a. oil of coconut ☒ b. oil of almonds
 c. oil of winter green d. oil of vitriol
- The role of H_2SO_4 in esterification reaction is.....
 a. oxidising agent b. reducing agent
☒ c. dehydrating agent d. hydrating agent
- The functional groups present in salicylic acid are
☒ a. carboxylic acid and alcoholic b. alcoholic and ester
 c. ester and amide d. carboxylic acid and amide
- Formation of water molecule in esterification reaction is by elimination of
 a. $-OH$ group from alcohol and 'H' from carboxylic acid
☒ b. $-OH$ group from carboxylic acid and 'H' from alcohol
 c. $-OH$ group from carboxylic acid and ' $-OH$ ' from alcohol
 d. 'H' from alcohol and ' $-OH$ ' from carboxylic acid

Short answer questions

1. What is Fischer esterification?

Fischer esterification is the reaction in which a carboxylic acid reacts with an alcohol in the presence of concentrated sulphuric acid as a catalyst to form an ester and water.

2. Write a general chemical equation showing formation of ester.



3. Write the role of conc. H_2SO_4 in esterification process?

Concentrated sulphuric acid acts as an acid catalyst and also as a dehydrating agent. It removes the water formed during the reaction and thus helps in increasing the yield of ester.

Remark and sign of teacher: